TRW Aeronautical Systems, Lucas Aerospace

MAINTENANCE MANUAL

23700
DC GENERATORS AND STARTER-GENERATORS

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Parts and assemblies not authorized for repair in the field must be sent to the TRW Aeronautical Systems, Lucas Aerospace Repair and Overhaul (R&O) at the location below..

REGION	ADDRESS	COMMUNICATIONS
ALL:	TRW Aeronautical Systems, Lucas Aerospace 777 Lena Drive Aurora, OH 44202 USA	PHONE (1) 330 995 1000 FAX (1) 330 995 1117

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RECORD OF REVISIONS

Revision Number	Date Issued	Date Inserted into CMM	Initials
Original Issue	Apr 15/68		
Revision 1	Mar 15/79		
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Retain this record in the front of the CMM. On receipt of revisions, insert affected pages into the manual. Record revision number, date issued, date inserted into the CMM, and initials on this page.

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RECORD OF TEMPORARY REVISIONS

Temporary Revision Number	Page Number	Date Issued	Date Inserted	Initials	Date Removed	Initials

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SERVICE BULLETIN LIST

SERVICE BULLETIN NUMBER	STATUS	REVISION NUMBER	DATE ISSUED	SERVICE BULLETIN NUMBER	STATUS	REVISION NUMBER	DATE ISSUED

Retain this record in the front of the CMM. On receipt of Service Bulletins, record the Service Bulletin number, status (valid or invalid), revision number (if applicable) and the date issued. Unless invalidated, all Service Bulletins affecting the 23700 DC Generators/Starter-Generators, should be kept with the CMM.

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SERVICE INFORMATION LETTER LIST

SERVICE INFORMATION LETTER NUMBER	STATUS	REVISION NUMBER	DATE ISSUED	SERVICE INFORMATION LETTER NUMBER	STATUS	REVISION NUMBER	DATE ISSUED

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INTRODUCTION

1. Purpose

This manual provides field maintenance procedures for TRW Aeronautical Systems, Lucas Aerospace generators and starter-generators. These procedures include the following:

- Generator/starter-generator removal from the aircraft
- Cleaning and resistance checking
- Brush removal
- Bench inspection
- Brush installation, seating, and run in
- Generator/starter-generator installation on the aircraft
- Packaging for shipment or storage
- Aircraft electrical system performance check

CAUTION: The maintenance and repair procedures performed by line maintenance personnel are restricted to those procedures found in this manual. Maintenance and repair procedures that are not part of this manual must be performed by an authorized FAA repair station.

You are advised to become thoroughly familiar with the equipment and the contents of this manual. Pay particular attention to applicable warnings and cautions. Before performing any procedure, become thoroughly familiar with the capabilities and limitations of the equipment. Make sure that all necessary safety equipment is on hand.

If an error, omission, or other technical discrepancy is found in this manual, fill out a copy of the Technical Publication Comment Form found at the back of the manual and forward it to the Customer Service Department at TRW Aeronautical Systems, Lucas Aerospace in Aurora, Ohio.

2. Product Support Insights

There are several different types and grades of aircraft brushes available from brush manufacturers. TRW Aeronautical Systems, Lucas Aerospace selects the best brush available that is appropriate for specific load and environmental conditions set forth by the airframe manufacturer. Due to variables in load conditions, altitude, generator cooling, and environmental conditions, a universal brush grade that meets all criteria of every aircraft application does not exist.

Brushes used in TRW Aeronautical Systems, Lucas Aerospace generators and starter-generators meet stringent performance requirements. Brush selection is decided upon considerable electrical design effort and extensive evaluation testing performed by TRW Aeronautical Systems, Lucas Aerospace and the airframe manufacturer during the aircraft certification test program. No matter what the application, all brushes must meet the following performance requirements and life objectives.

- The brushes must operate up to 100% of rated generator load under all specified environmental conditions. This includes self-cooled operation at sea level, all operating altitudes, and the maximum aircraft certification altitude. The brushes must have the ability to handle the generator overload operation that may occur during battery charging and system fault conditions.
- The brushes must operate under specific generator cooling air conditions which are often restrictive. The brushes must be capable of withstanding the operating temperature resulting from normal and aborted engine starting or engine wash/rinse cycles using both onboard battery and external ground power unit supplies.
- The brushes must be able to provide an acceptable level of brush life under all aircraft operating conditions. Typical acceptable brush life levels vary from 400 to over 1500 hours, even up to 2000 for some applications, depending on the aircraft installation, operating and cooling conditions.
- The brushes must demonstrate an acceptable level of commutator life under all aircraft operating conditions. Typical commutator life ranges vary from 4,500 to over 80,000 hours, depending on the aircraft installation, operating and cooling conditions, and the grade of brush.
- The brushes must provide acceptable commutation, with minimum ripple voltage and electromagnetic interference (EMI).

A. Commutator Filming

Brush performance is directly dependent on the formation of a micro-thin oxide “film” which develops on the surface of the commutator during operation. This film is the result of sliding and electrical conduction. The oxide formation relies on the presence of water vapor and oxygen, or on the introduction of lubricating additives in the brush to provide satisfactory filming at all altitudes. Filming is not constant and depend on load and environmental conditions.

B. TRW Aeronautical Systems, Lucas Aerospace Furnishes Two Types of Brushes

(1) Carbon graphite

Carbon graphite brushes are a combination of lamp black, petroleum coke, artificial graphite, and an additive dispersed throughout the brush. The additive used for aircraft applications is normally molybdenum disulfide (MoS_2). MoS_2 compensates for the lack of moisture during altitude operation and maintains commutator filming. The additive is processed with carbon graphite materials prior to being fired at approximately 1100° C. This brush is commonly referred to as a particle flake or dispersed moly type of brush.

(2) Electrographic

Electrographic brushes are composed of the same materials as the carbon graphite grade, but are processed at a firing temperature of approximately 2000° C. The additive, or additives, are not dispersed through the brushes because it would be destroyed at the higher firing temperature. The additives are added to the brush after it is formed.

(a) Core-type brushes are manufactured with MoS_2 tapped into small diameter holes in the brush.

(b) Another type of core-type brush is manufactured with additives added by vacuum pressure impregnation (VPI), in addition to the MoS_2 cores.

(c) Long-life brushes are non-core type electrographic brushes which have additives added to the brushes by VPI.

C. Brush and commutator wear

Brush and commutator wear is a function of electrical load and environmental conditions and the brush grade. During brush run-in, a lighter load is recommended to avoid arcing and possible burning of the commutator. Operation at a higher speed increases the rate at which a brush will wear and seat. Seating in the generator mode is recommended rather then the motoring mode.

The surface finish of the commutator influences the rate at which the brush seats. Surface finishes less than 32 micro-inches (0.8 micro-meters) tend to result in long brush seating operations. Surface finishes much less than 32 micro-inches (0.8 micro-meters) tend to glaze the commutator. Surface finishes greater than 120 micro-inches (3 micro-meters) may produce excessive initial brush wear. During field operations, individual brush wear rates may vary significantly depending on the brush grade, load, and environmental conditions.

The main cause of non-uniform brush wear within a unit is selectivity, i.e. the tendency for a higher percentage of the total current to flow in one brush. In every four-pole generator, there are four parallel paths for the current to flow. The resistance of each path determines the percentage of load current that flows through each path. A unit which has been subjected to heavy loads will tend to experience greater signs of selectivity.

Another factor of non-uniform brush wear is dependent on whether or not a brush has cores. Brushes with cores tend to form grooves in the commutator after running for a prolonged time. This is due to the abrasive action of the core material, which is generally different than that of the base material, and varies according to the brush and application.

Significant variations in spring pressure can also influence brush and commutator wear characteristics. Too light of a spring pressure will result in high wear rates due to electrical wear. Too heavy of a spring pressure will also result in high wear due to the increased friction between the brush face and commutator.

3. Time Between Overhaul (TBO)

CAUTION: Do not exceed 1000 hours TBO unless the aircraft maintenance or flight manual specifies a different TBO.

DC brush type generators and starter-generators must undergo periodic inspection between overhauls. The frequency of inspection is generally specified by the aircraft manufacturer or by an operator's field service experience. If an inspection frequency is not specified, a sampling inspection period of 250 hours is recommended by TRW Aeronautical Systems, Lucas Aerospace.

4. Safety Advisory

This manual describes physical and chemical processes which require the use of chemical or other commercially available materials that require precautionary attention.

The user of this manual should obtain material safety data sheets (MSDS), Occupational Safety and Health Act (OSHA) Form 20 or equivalent, from the manufacturers or suppliers of materials that are specified in the materials list in this manual. Users of this manual are also advised to refer to applicable safety information contained in the "NIOSH Occupational Guideline for Chemical Hazards": published by the United States Department of Labor.

Warnings and Cautions are defined as follows:

WARNING: **ALERTS OPERATING AND MAINTENANCE PERSONNEL TO POTENTIAL HAZARDS THAT COULD RESULT IN PERSONAL INJURY. WARNINGS DO NOT REPLACE THE MANUFACTURER'S RECOMMENDATIONS.**

CAUTION: Alert operating and maintenance personnel to conditions that could result in equipment damage.

5. Materials List

A warning and/or caution will precede the use of the materials listed in Table 1.

WARNING: **BEFORE USING ANY OF THE FOLLOWING MATERIALS, BE AWARE OF ALL HANDLING, STORAGE AND DISPOSAL PRECAUTIONS RECOMMENDED BY THE MANUFACTURER OR SUPPLIER. FAILURE TO COMPLY WITH THESE RECOMMENDATIONS CAN RESULT IN SERIOUS INJURY, PHYSICAL DISORDER, OR DEATH.**

Material	Used in ...
Compressed Air	Bench Inspection
Isopropyl Alcohol	Bench Inspection and Brush Seating and Run-in
Lubricating Oil	Brush Seating and Run-in

Table 1 - Hazardous Materials

6. Non TRW Aeronautical Systems, Lucas Aerospace Authorized Components and Processes Policy

TRW Aeronautical Systems, Lucas Aerospace authorizes the use of genuine TRW Aeronautical Systems, Lucas Aerospace spare parts which meet stringent engineering design specifications and quality standards, and have traceability to having been procured and certified to these specifications by the Lucas Aerospace Quality Assurance incoming and in-process inspection systems. The Lucas Aerospace Customer Service Centers are the only authorized distributors of genuine TRW Aeronautical Systems, Lucas Aerospace replacement parts and complete units.

All repair and overhaul facilities are obligated to provide the FAA, or any other in-county air authority, with proper traceability documentation indicating approval of all spare parts, materials and processes to ensure configuration compliance and continued air worthiness.

The use of any non-TRW Aeronautical Systems, Lucas Aerospace unauthorized parts, or any parts not having been submitted to the Lucas Aerospace Quality Assurance inspection system will invalidate any and all factory warranties. All TRW Aeronautical Systems, Lucas Aerospace warranties are automatically voided on any TRW Aeronautical Systems, Lucas Aerospace designed unit that has been modified by the installation of any unauthorized parts, materials or unapproved processes supplied by other outside services. The repair station's quality assurance activity shall assume product liability for all units that have been modified in this fashion.

Damage resulting from the use of non-TRW Aeronautical Systems, Lucas Aerospace replacement parts, materials or processes is not covered by the TRW Aeronautical Systems, Lucas Aerospace warranty or service policy for any product or for any application.

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DESCRIPTION AND OPERATION

1. General

TRW Aeronautical Systems, Lucas Aerospace generators and starter-generators provide a continuous 30 VDC power supply to an aircraft when driven by the engine or gearbox. Starter- generators have the additional function of acting as a starter motor during the engine start cycle.

Generators and starter-generators both function in a power generation system with a generator control unit (GCU). The GCU provides the regulation, control, and protective functions required for overall electrical system operation.

2. Generating Mode

By continuously adjusting the generator's field excitation, the GCU maintains a constant point of regulation (POR) voltage over varying generator operating conditions. This control provides for regulated output power at generator terminal B while the generator or starter-generator is operating in the generator mode.

3. Starter-Generator Start Cycle Operation

TRW Aeronautical Systems, Lucas Aerospace starter-generators are designed to operate as series or shunt motors during the engine start cycle.

A. Series Starters

The series starter design incorporates shunt field, interpole, and compensating windings. An additional start winding is connected in series with the armature. The start winding is connected to terminal C on the terminal block. This terminal serves as the input connection point for the engine starting power source. The series starter provides the maximum starting torque and the maximum no-load speed.

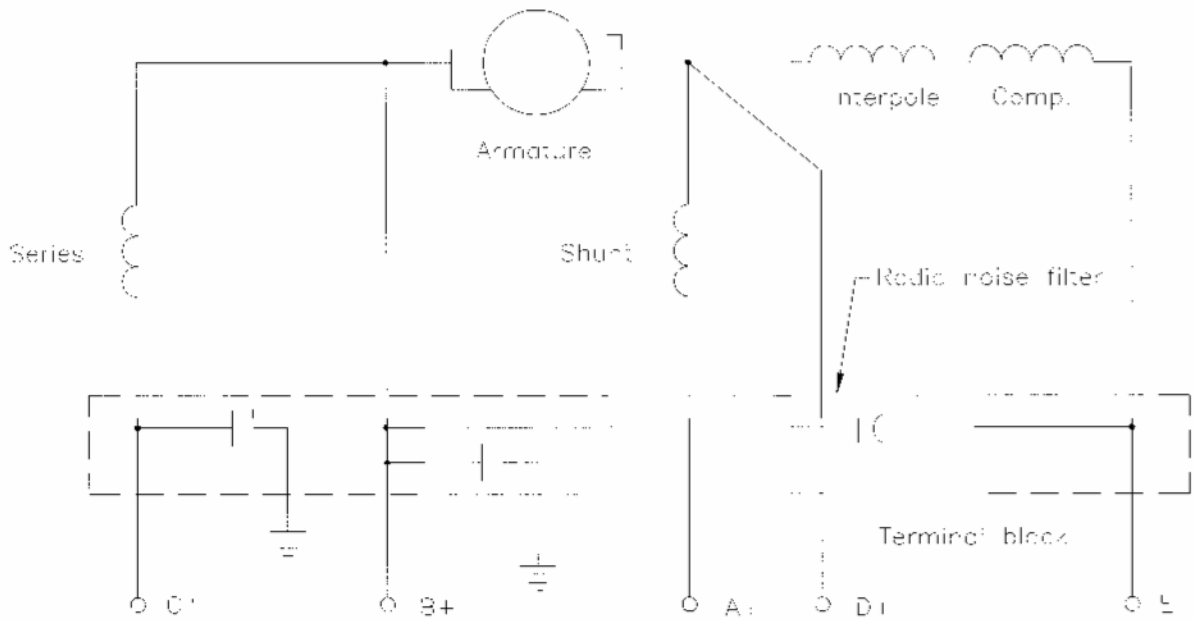

Figure 1 - Series Start Schematic Diagram

B. Shunt Starters

The shunt starter design incorporates shunt field, interpole, and compensating windings. It does not have an additional start winding like the series start design. Terminal B on the terminal block serves as the input connection point for the engine starting power source. After the engine has started, terminal B serves as the generator's output terminal.

The required starter field excitation current is supplied through terminal A. The shunt field excitation current is controlled by circuitry within the GCU during the engine start cycle. This circuitry, commonly referred to as field weakening control, monitors the interpole voltage (terminals D and E) to determine the starter's armature current. When the armature current falls below a pre-set level (usually the full rated output current of the generator) the shunt field excitation is reduced. This high armature current results in a high starting torque. Field weakening control is active during the high speed portion of the engine start cycle. The shunt start design provides higher mid-range and high speed starting torque.

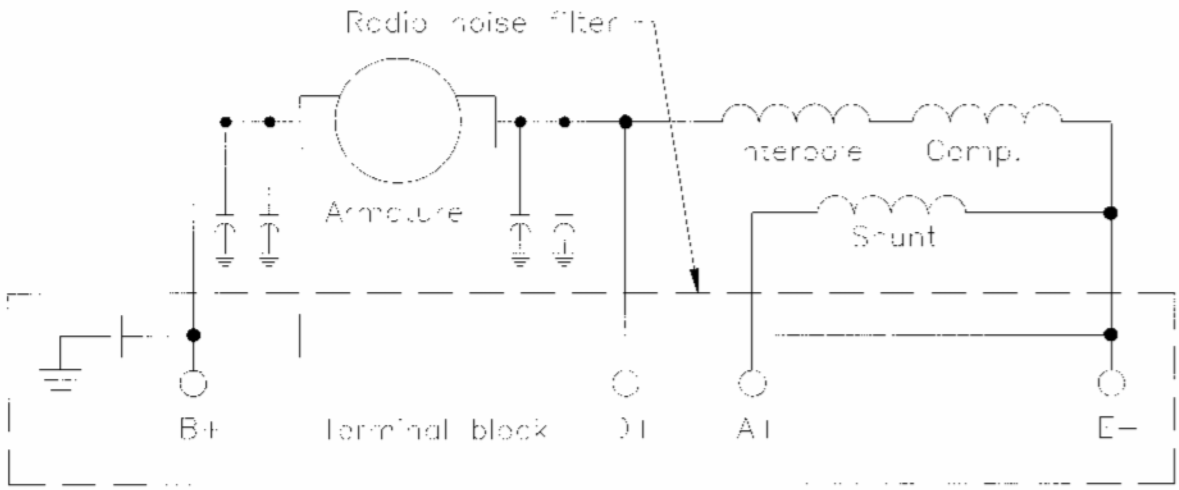

Figure 2 - Shunt Start Schematic Diagram

C. Hybrid Configuration

The hybrid design is most similar to the series starter design with the exception that the start winding is connected in series with the armature through generator terminal B. This design has the high starting torque and high no-load speed characteristics of the series starter design, however, its rated output is obtained at a somewhat higher minimum speed. In this design, terminal B serves the same two functions as the shunt configuration.

4. Cooling

The starter-generator is cooled by either blast (ram) air, an integral fan mounted on the armature shaft, or a combination of both.

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BENCH INSPECTION

1. Introduction

This section provides the disassembly, field cleaning, resistance check and bench inspection procedures for DC Generators and Starter-Generators.

2. Tools and Materials

The following tools and materials are required to perform a bench inspection of DC Generators and Starter-Generators.

Tools/Materials Description	Description/Specification	Source
Brush	Soft Bristle, non-metallic	Commercially Available
Cleaning pads/wiping cloths	Lint-free, soft fabric	Commercially Available
Compressed Air	Not applicable	Not applicable
Isopropyl Alcohol	TT-I-735, Grade A Read **WARNING** before using. See MSD Sheet.	Commercially Available
Ohmmeter or low voltage bridge	50 VDC max.	Commercially Available
Stator Support Fixture	Fabricate from Figure 5001	Not applicable
Wire Hook Tool	Not applicable	Commercially Available

Table 2 - Inspection Tools/Materials

3. Remove the Unit from the Aircraft

Brush inspection, brush change, and cleaning can be accomplished only when the unit is removed from the aircraft.

A. Remove the terminal block cover (where applicable)

Remove the attaching hardware that attaches the terminal block cover. In some cases, pull the terminal block cover off of the terminal block.

B. Identify all electrical leads

(1) Place identifications tags on any unmarked electrical leads.

(2) Disconnect the leads at the terminal block and connector (if applicable).

C. Remove the unit from the aircraft

NOTE: Procedures for three common attachment style are outlined below.

CAUTION: The unit must be supported at all times during removal from the aircraft. Do not allow the unit to hang unsupported unless the V-band or the attaching hardware is fully tightened. Excessive bending loads on the drive shaft shear section can damage the unit.

(1) Quick Attach-Detach (QAD) Kit removal

(a) Remove the duct from the Air Inlet or Fan Cover.

(b) Loosen the nut securing the V-band Clamp. Make sure to support the unit.

(c) Unlatch the T-bolt and release the V-band.

(d) Carefully remove the unit from the spline drive.

(2) Slotted mounting holes (button holes)

(a) Remove the duct from the Air Inlet or Fan Cover.

(b) Loosen the attaching hardware securing the unit to the accessory mounting pad. Make sure to support the unit.

(c) Rotate the unit until the clearance holes align with the attaching hardware.

(d) Carefully remove the unit from the spline shaft.

(3) Round mounting holes.

(a) Remove the duct from the Air Inlet or Fan Cover.

(b) Loosen the attaching hardware securing the unit to the accessory mounting pad. Make sure to support the unit.

(c) Remove the attaching hardware.

(d) Carefully remove the unit from the spline shaft.

D. Inspect the air inlet and exhaust screens (if applicable) for Foreign Objects and Debris (FOD)

Remove any FOD and check for internal damage.

4. Field Cleaning and Resistance Check

This section applies to those units which require new brushes. The brushes must remain installed in the unit during the internal cleaning and resistance check procedures.

A. Remove the brush access cover (or in some cases where a brush access cover is not part of the unit, remove the air inlet/fan cover).

(1) Brush access cover

Loosen the screw that secures the brush access cover around the unit. Remove the brush access cover from the unit.

(2) Air Inlet

(a) Remove the screws that secure the air inlet/fan cover to the anti-drive end end bell assembly.

(b) Remove the air inlet/fan cover from the unit.

B. Clean the interior and exterior of the unit

WARNING: COMPRESSED AIR USED FOR CLEANING CAN CREATE AIRBORNE PARTICLES THAT MAY ENTER THE EYES. PRESSURE MUST NOT EXCEED 30 PSIG (207 kPa). EYE PROTECTION IS REQUIRED.

CAUTION: Do not direct the air jet at the bearings to avoid damage to the bearing seals.

CAUTION: Be sure Compressed Air is free of oil, water or any other contaminates.

(1) Remove carbon and copper dust from the armature an stator windings using compressed air. Start at the drive end and work towards the anti-drive end of the unit.

WARNING: ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. DO NOT USE NEAR OPEN FLAMES, WELDING AREAS, OR ON HOT SURFACES. INHALATION OF VAPORS CAN CAUSE, DIZZINESS, DROWSINESS AND HEADACHE. CONTACT WITH SKIN CAN CAUSE IRRITATION. IF HANDLING LARGE QUANTITIES [1 GALLON (3.79 L) OR MORE], USE IN A WELL-VENTILATED AREA

(2) Remove oil and/or contaminants from the interior of the unit with a soft, lint-free cloth moistened with isopropyl alcohol.

(3) Thoroughly clean the exterior of the unit using a soft cleaning cloth and isopropyl alcohol.

(4) Wipe the exterior of the unit dry with a clean, lint-free cloth.

C. Resistance Check

CAUTION: To avoid possible damage to the radio interference parts, do not hi-pot (including the use of a megohmeter) unless instructed to do so in the Component Maintenance Manual (CMM) for the unit under test.

Check the DC Resistance between positive terminal B and the machine frame using an ohmmeter or low voltage bridge.

(1) The resistance must measure 10,000 ohms or more. If the resistance measures less than 10,000 oms, clean and test the unit again.

(2) If the unit fails the resistance check, it must be repaired or overhauled. Package and ship the unit to a repair station, authorized for electrical accessory overhaul. Include a brief description of the damage.

5. Bench Inspection

The following bench inspection is required for all units during brush inspection. Inspect the following components of the generator or starter-generator.

NOTE: If a damage condition exists for which no corrective action is outlined in this manual, install the original brushes and brush access cover or air inlet/fan cover (as applicable). **Do not install the damaged unit on the aircraft.** Package and ship the unit to a repair station, authorized for electrical accessory overhaul. Include a brief description of the damage.

CAUTION: The maintenance and repair of the unit by line maintenance personnel is limited to the procedures detailed in this manual. Maintenance and repair procedures not found in this manual must be performed by an FAA repair station authorized for accessory overhaul.

A. Terminal Block

(1) Visually examine all surfaces for cracks.

(2) Visually examine the terminal studs for crossed or stripped threads.

(3) Visually examine the terminal studs for burning, melting, or discoloration.

(4) If damage is found, the unit must be repaired or overhauled.

(5) Package and ship the unit to a repair station, authorized for electrical accessory overhaul. Include a brief description of the damage.

B. Connector and spacer (for units with a speed pickup)

(1) Visually examine all exposed surfaces of the connector and spacer for cracks and other apparent physical damage.

(2) Make sure that the connector pins are not bent or missing.

(3) If damage is found, the unit must be repaired or overhauled.

(4) Package and ship the unit to a repair station, authorized for electrical accessory overhaul. Include a brief description of the damage.

C. Drive End End Bell Assembly

(1) Visually examine all surfaces for cracks.

(2) Make sure all attaching hardware is secure.

(3) Visually examine the mating surface for fretting corrosion.

(4) Visually examine the screen (if applicable) for tears.

(5) If damage is found, the unit must be repaired or overhauled.

(6) Package and ship the unit to a repair station, authorized for electrical accessory overhaul. Include a brief description of the damage.

D. Speed Pickup, Guard, and Grommet (if applicable)

(1) Visually examine the speed pickup and guard for damage.

(2) Be sure the hardware attaching the guard to the drive end end bell is secure.

(3) Be sure the speed pickup jam nut is secure.

(4) Visually examine the grommet for cracking or dryness. Be sure the speed pickup leads are routed through the grommet.

(5) Be sure the insulation sleeving covering the speed pickup leads is in place.

(6) Visually examine the insulation sleeving for tears and cracks.

(7) If damage is found, the unit must be repaired or overhauled.

(8) Package and ship the unit to a repair station, authorized for electrical accessory overhaul. Include a brief description of the damage.

E. O-ring (if applicable)

(1) Visually examine the O-ring for cracks, cuts, dryness, or wear.

(2) If damage is found, refer to the applicable CMM for the replacement part number.

(3) Replace the O-ring.

F. Drive Shaft (See Figure 3)

NOTE: Do not disassemble the drive shaft and dampener assembly unless damage is suspected.

(1) Visually examine exposed surfaces for wear.

(2) Visually examine the drive spline for rounding, stripping, or uneven wear.

(3) If damage is found, the unit must be repaired or overhauled.

(4) Package and ship the unit to a repair station, authorized for electrical accessory overhaul. Include a brief description of the damage.

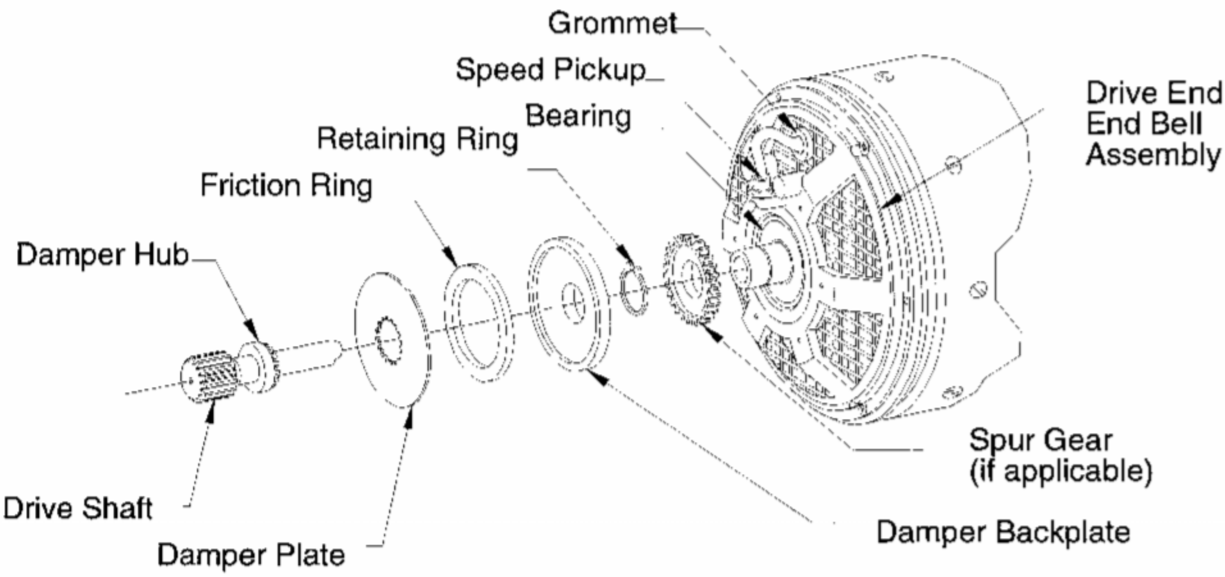

Figure 3 - Drive End End Bell Assembly, Shaft, Dampener Assy., etc.

G. Dampener Assembly (See Figure 1001)

NOTE: Do not disassemble the drive shaft and dampener assembly unless damage is suspected.

(1) Visually examine all exposed surfaces for cracks, looseness of parts or other apparent physical damage.

(2) Visually examine the friction ring.

(3) If damage is found, the unit must be repaired or overhauled.

(4) Package and ship the unit to a repair station, authorized for electrical accessory overhaul. Include a brief description of the damage.

H. Air Inlet/Fan Cover

NOTE: Figure 1002 illustrates two of many different types of air inlets used on TRW Aeronautical Systems, Lucas Aerospace generators and starter-generators.

(1) Visually examine all surfaces for cracks.

(2) Visually examine all surfaces for dents, warping, nicks, and scratches that reduce the clearance between the sidewall and the fan.

NOTE: Continue the inspection only if the air inlet/fan cover was removed from the unit.

(3) Visually examine the mounting and inlet flanges for gouging, scoring, or glazing.

(4) Visually examine the insulating tape (if applicable) for tears or frayed edges.

(5) Visually examine the screen (if applicable) for tears

(6) If damage is found, the unit must be repaired or overhauled.

(7) Package and ship the unit to a repair station, authorized for electrical accessory overhaul. Include a brief description of the damage.

I. Fan

(1) Visually examine all exposed surfaces for cracks.

(2) Visually examine all blades for warping, nicks and scratches.

(3) If damage is found, the unit must be repaired or overhauled.

(4) Package and ship the unit to a repair station, authorized for electrical accessory overhaul. Include a brief description of the damage.

J. Brush Access Cover (if applicable)

(1) Visually examine all surfaces for cracks.

(2) Visually examine all surfaces for dents and warping.

(3) Visually examine the insulating tape (if applicable) for tears or frayed edges

(4) If damage is found, the unit must be repaired or overhauled.

(5) Re-install any removed parts.

(6) Package and ship the unit to a repair station, authorized for electrical accessory overhaul. Include a brief description of the damage.

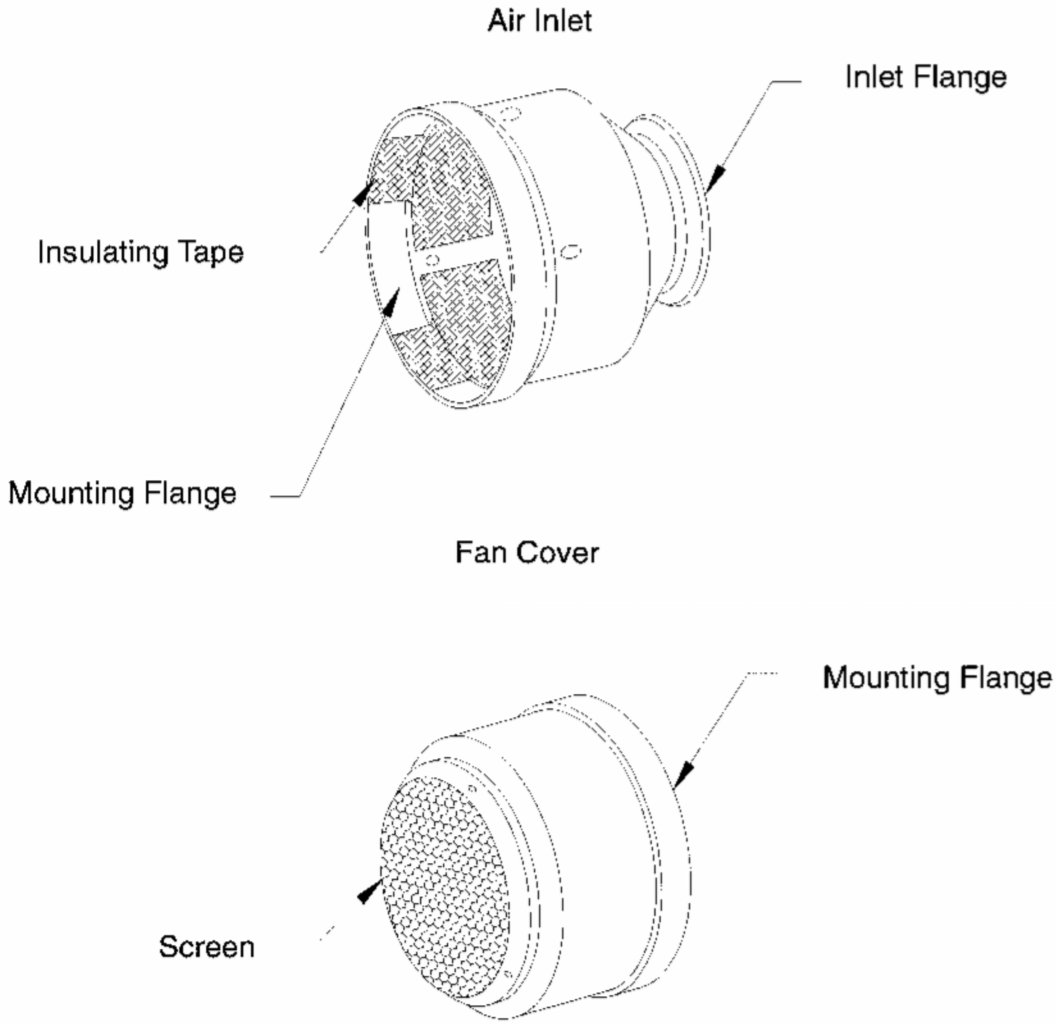

Figure 4 - Typical Air Inlet/Fan Covers

K. Brushes

(1) Identify the brush holder assemblies with the numbers 1 through 4 as shown in Figure 5.

(2) Remove the screws that attach the brush shunt leads to the brush holder assemblies.

CAUTION: Do not lift the brush springs more than is needed to remove the brush sets from the brush holders. Overstressing the brush springs can reduce spring pressure.

(3) Use a wire hook to lift the brush spring away from the brush set.

(4) Remove the brush set from the brush holder and identify the brush set with the corresponding number on the brush holder.

(5) Slowly return the brush spring to a resting position.

(6) Repeat steps (3) through (5) at the remaining brush holder locations.

(7) Visually examine the brushes for cracks, chips, frayed leads, loose rivets, and loose shunt connections.

NOTE: Brush replacement should be accomplished only after the completion of all bench inspections.

(8) Estimate the remaining life of each brush by comparing the wear line with the illustrations in Figure 6.

(9) If the remaining life of any brush is less than the next service interval, replace all the brushes as a set. If any brush is damaged, replace all brushes as a set.

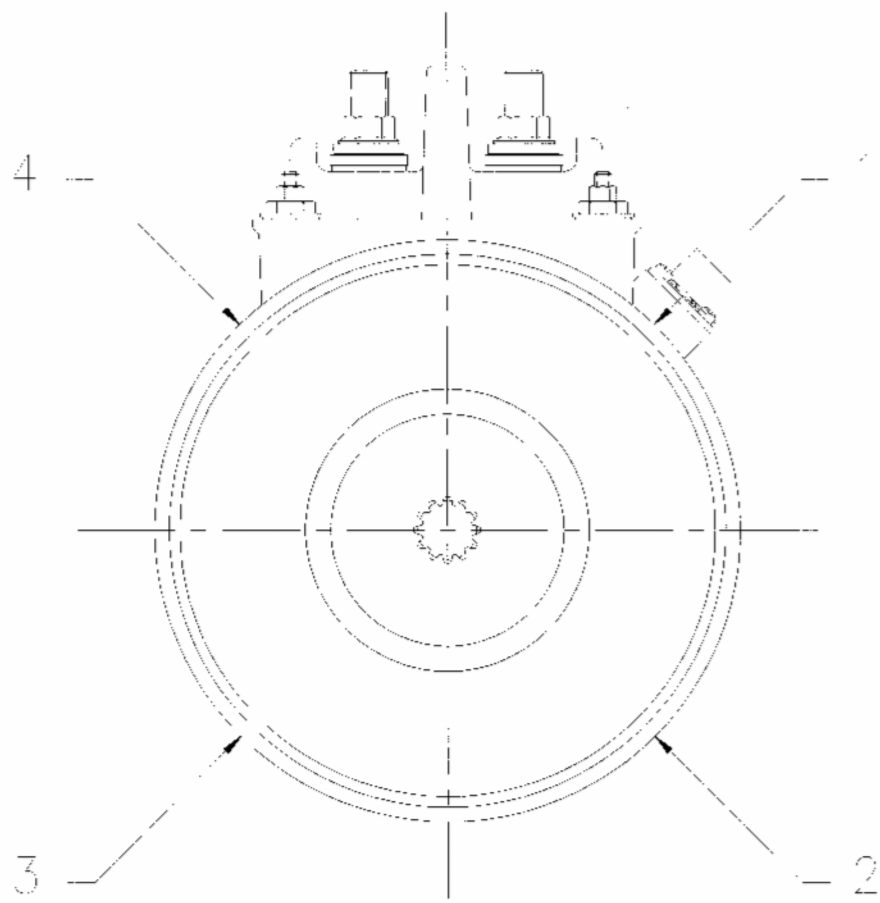

Figure 5 - Brush Numbering Viewed from the Drive End

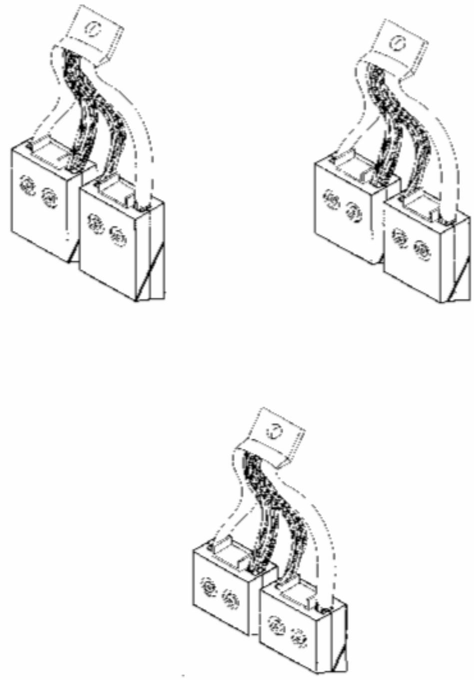

Figure 6 - Brush Wear

L. Bearings

(1) Put the generator or starter-generator drive end down on a vertical stator support.

<u>NOTE:</u> Some bearings use a high temperature lubricant which will cause a good bearing to feel rough.

(2) Turn the fan slowly in the normal direction of rotation to obtain the "feel" of the bearings. The rotation should be relatively smooth and the bearings should not sound dry.

(3) If damage is found, the unit must be repaired or overhauled.

(4) Re-install any removed parts.

(5) Package and ship the unit to a repair station, authorized for electrical accessory overhaul. Include a brief description of the damage.

M. Anti-drive End End Bell Assembly

(1) Visually examine all exposed surfaces for cracks.

(2) Hand tighten a machine screw into the blind rivet nut in the brush holder assembly to make sure that the self-locking feature is functional. The machine screw should bind in the blind rivet nut before it is fully engaged. (See Figure 7).

(3) Visually examine the brush holders, including the brush spring supports, for cracks or warping.

(4) Visually examine the brush holders, including the center support, for physical damage such as discoloration or burning caused by electrical arcing.

(5) Visually examine the brush springs for cracks, warping, discoloration or melting.

(6) Be sure the hardware that attaches the brush holders to the anti-drive end end bell is secure.

(7) Be sure the hardware that attaches the anti-drive end end bell to the stator and housing assembly is secure.

(8) If damage is found, the unit must be repaired or overhauled.

(9) Re-install any removed parts.

(10) Package and ship the unit to a repair station, authorized for electrical accessory overhaul. Include a brief description of the damage.

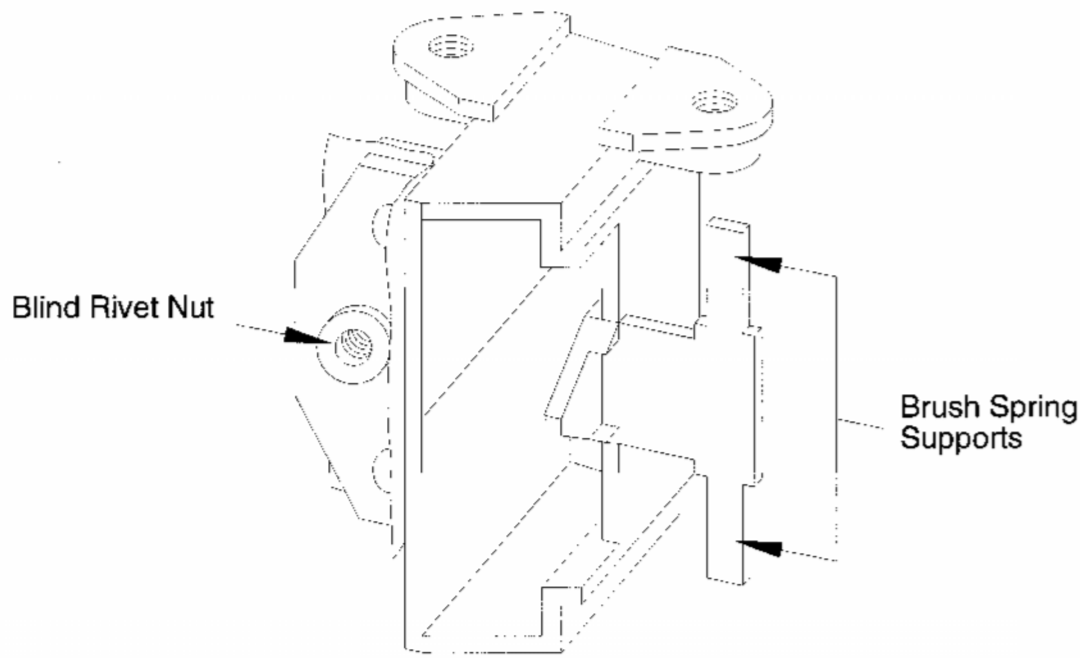

Figure 7 - Inspecting the Brush Holder Assembly

N. Commutator

(1) Visually examine all exposed surfaces for cracks.

(2) Visually examine the commutator bars for burning or discoloration.

The commutator should have a burnished appearance with light filming. A defective commutator is black, burned, high barred, or pitted.

(3) If the brushes need to be replaced, determine the remaining mica undercut.

The remaining mica undercut depth must be 0.20 inches (0.51 mm) or greater.

NOTE: The use of electrographitic cored brushes will form grooves on the commutator.

(4) Calculate the depth of groves by measuring the corresponding protrusions on the brushes. (See Figure 8)

The groove depth in the commutator must be 0.20 inches (0.51 mm) or less.

NOTE: Because this publication applies to so many different models, the groove depth described above may be conservative in some cases.

(5) If damage is found, the unit must be repaired or overhauled.

(6) Re-install any removed parts.

(7) Package and ship the unit to a repair station, authorized for electrical accessory overhaul. Include a brief description of the damage.

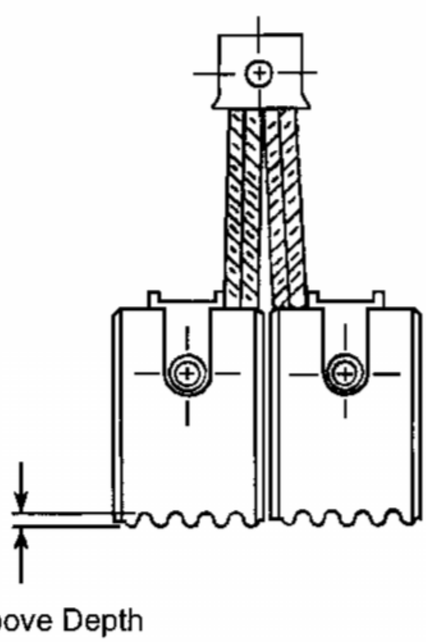

Figure 8 - Calculating Commutator Groove Depth

O. Brush Installation after Inspection

(1) Set the unit, drive end down, on a vertical stator support fixture.

NOTE: If the original brushes are found to be acceptable during bench testing and will be used in the unit, it is very important that the numbers marked the brushes are matched with the numbers on the brush holder assemblies and installed into their original positions. (See Figure 3)

(2) At each brush holder location, lift one brush spring at a time with a wire hook tool and insert the matching original brush set into the brush holder assembly.

(3) Slowly lower the brush spring into position on the brush set.

(4) Attach each stator harness strap and brush lead to the nearest brush holder assembly with a screw.

(5) Press the brush leads over the brush springs. Be sure the brush leads do not get caught under the brush springs

NOTE: Depending on the brush configuration, the brush leads may or may not have insulation sleeving.

CAUTION: 150 and 200 Amp units: During brush removal and installation, it may be possible for the inboard brush springs to be positioned closer to the anti-drive end end bell after brush installation. (See Figure 9)

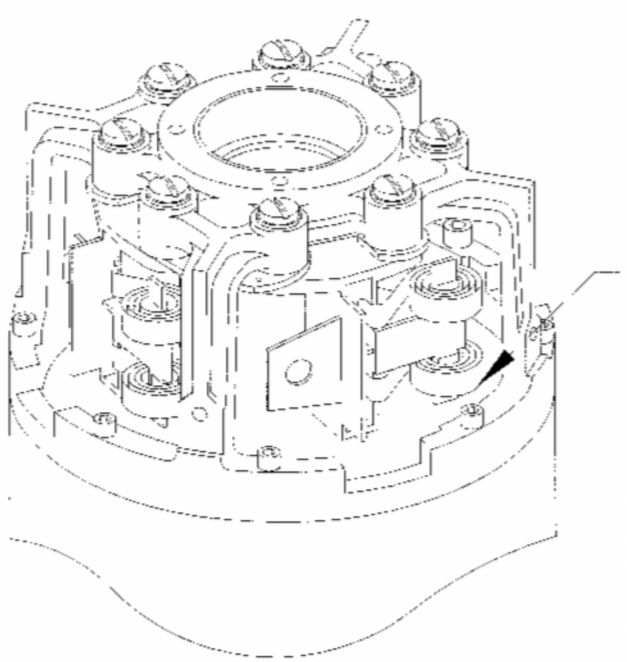

Figure 9 - 150 and 200 Amp Units

(6) Be sure the brush springs are properly in position and away from the anti-drive end end bell after brush insulation. (See Figure 10)

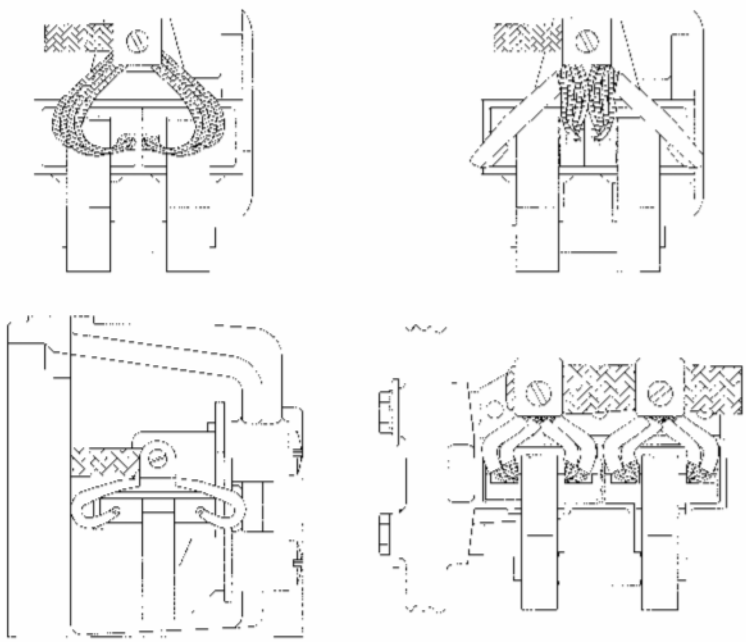

Figure 10 - Brush Lead Placement After Installation

P. V-band Clamp (if applicable)

(1) Visually examine all surfaces for cracks.

(2) Visually examine the self-locking nut and T-bolt for crossed or stripped threads.

(3) Replace the T-bolt or self-locking nut if damaged. Replace V-band if damage is excessive.

Q. Mounting Adapter (if applicable)

(1) Visually examine all surfaces for cracks.

(2) Visually examine the mating surfaces for fretting.

(3) Visually examine the mating surfaces for gouging, scoring or glazing.

(4) Replace the mounting adapter if excessive damage is found.

BRUSH INSTALLATION, SEATING AND RUN-IN

NOTE: If bench inspection reveals the original brushes are acceptable, the brush run-in procedure (paragraph 3 of this section) is not required. If new brush sets are used, brush seating and run-in is required.

NOTE: If functional faults or material defects are found during bench testing, the unit must be repaired or overhauled. Install all removed items back into the unit. Package and ship the unit to a repair station authorized for electrical accessory overhaul. Include a brief description of the damage.

1. Tools and Materials

The following tools and materials are required for proper brush installation, seating and run-in.

Tool/Material	Description/Specification	Source
Compressed Air	Not applicable	Not applicable
Isopropyl Alcohol	TT-I-735, Grade A Read the **WARNING** and MSDS before using this material	Commercially Available
Lubricating Oil	MIL-L-23699	Commercially Available
Masking Tape	Not Applicable	Commercially Available
Sandpaper	180 grit, non-aluminum oxide	Commercially Available
Stator Support Fixture	Make from Figure 5001	Not applicable
Spline Wrench	See applicable CMM	Not applicable
Tweezers	Not applicable	Commercially Available
Wire Hook Tool	Not applicable	Commercially Available

Table 3 - Tools and Materials

2. Brush Installation and Seating

NOTE: All generators and starter-generators that had a brush change must be seated and run-in IAW the instructions in this section before returning the unit to service.

A. Prepare the generator or starter-generator for brush run-in

(1) Cut a strip of 180 grit sandpaper, slightly wider than the brush contact area and one inch longer than the circumference of the commutator.

WARNING: **ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. DO NOT USE NEAR OPEN FLAMES, WELDING AREAS, OR ON HOT SURFACES. INHALATION OF VAPORS CAN CAUSE, DIZZINESS, DROWSINESS AND HEADACHE. CONTACT WITH SKIN CAN CAUSE IRRITATION. IF HANDLING LARGE QUANTITIES [1 GALLON (3.79 L) OR MORE], USE IN A WELL-VENTILATED AREA**

(2) Clean any four adjacent commutator bars with isopropyl alcohol.

NOTE: Make sure that the taped end of the sandpaper is in the normal direction of rotation and that the abrasive side of the sandpaper faces away from the commutator.

(3) Tape the leading edge of the sandpaper to the clean commutator bars with masking tape. (Ref Figure 11)

(4) Carefully turn the drive shaft in the direction of rotation with a spline wrench so the sandpaper strip wraps around the commutator until the trailing edge of the sandpaper covers the masking tape.

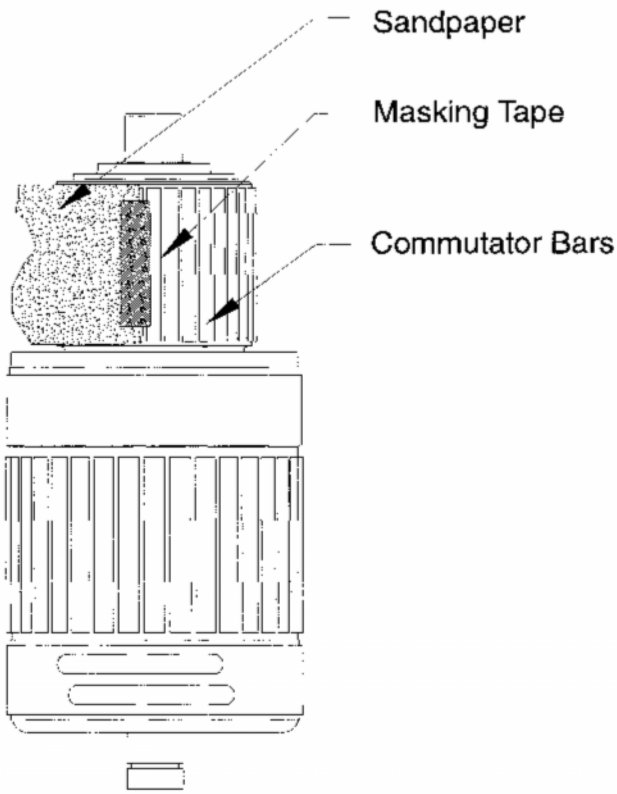

Figure 11 - Attaching the Sandpaper

B. Insert the Brushes

(1) Set the unit, drive end down, on a vertical stator support fixture.

CAUTION: Raise and lower the brush springs slowly. Brush springs can be damaged if the are allowed to snap back into position.

(2) At each brush holder location, lift one brush spring at a time with a wire hook tool and insert a new brush set into the brush holder assembly.

(3) Slowly lower the brush spring into position on the brush set.

NOTE: The sandpaper should contact the leading edge (shorter side) of the brush first when the armature is rotated in its normal direction.

(4) Carefully turn the drive shaft in the direction of rotation with a spline wrench. Make 2-3 complete turns of the drive shaft.

NOTE: The sanded portion of the brush will appear dark and rough in comparison to an unseated brush. A coarsely seated brush has 100% of the contact surface sanded.

(5) Remove one brush as a sample. If the brush is not pre-seated on 100% of the contact area, reinstall the brush into its original location and turn the armature one complete turn. Continue this procedure until all brushes are coarsely seated.

(6) Carefully raise each brush spring and move the brushes about half-way out of the brush holders. Gently place the brush spring against the side of the brush to hold it away from the commutator surface.

C. Remove the sandpaper and Blow out the Carbon Residue from Around the Commutator

(1) Grasp the trailing edge of the sandpaper with a pair of tweezers through an opening in the anti-drive end end bell assembly.

(2) Pull the sandpaper through the opening. Detach the masking tape.

WARNING: **COMPRESSED AIR USED FOR CLEANING CAN CREATE AIRBORNE PARTICLES THAT MAY ENTER THE EYES. PRESSURE MUST NOT EXCEED 30 PSIG (207 kPa). EYE PROTECTION IS REQUIRED.**

CAUTION: Do not direct the air jet at the bearings to avoid damage to the bearing seals.

CAUTION: Be sure Compressed Air is free of oil, water or any other contaminates.

(3) Blow out the carbon residue from inside unit using dry and clean compressed air.

WARNING: ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. DO NOT USE NEAR OPEN FLAMES, WELDING AREAS, OR ON HOT SURFACES. INHALATION OF VAPORS CAN CAUSE DIZZINESS, DROWSINESS AND HEADACHE. CONTACT WITH SKIN CAN CAUSE IRRITATION. IF HANDLING LARGE QUANTITIES [1 GALLON (3.79 L) OR MORE], USE IN A WELL-VENTILATED AREA

(4) Remove any residue left from the masking tape on the commutator bars with isopropyl alcohol.

(5) Lower each brush set into its brush holder assembly.

(6) Attach each stator harness strap and brush lead to the nearest brush holder assembly with a screw (Ref Figure 12).

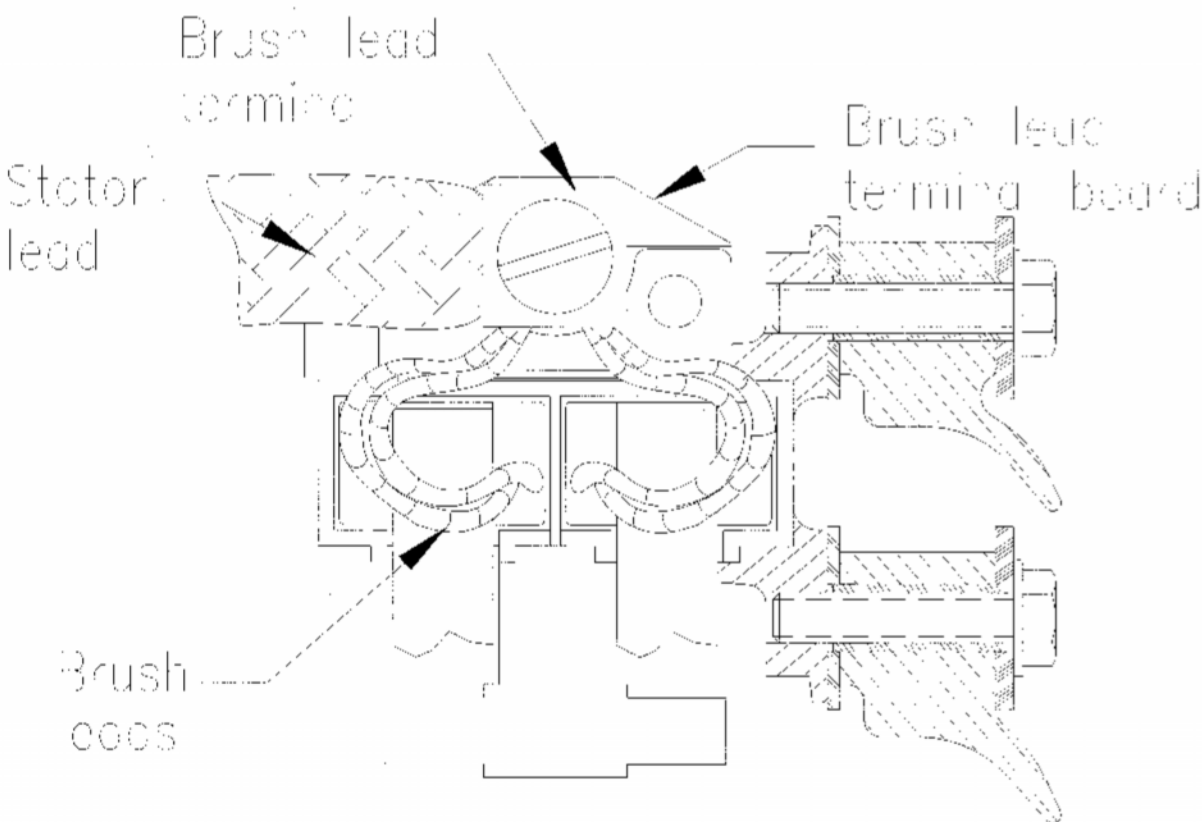

Figure 12 - Brush Lead Connections

(7) Press the brush leads over the brush springs. Make sure that the brush leads do not get caught under the brush springs.

CAUTION: 150 and 200 A units: During brush removal and installation, it may be possible for the inboard brush springs to be positioned closer to the anti-drive end end bell after brush installation. (Ref Figure 9)

(8) Be sure the brush springs are properly in position and away from the anti-drive end end bell after brush insulation. (Ref Figure 10)

3. Final Brush Run-in

This paragraph provides instructions for final brush run-in. Two common methods are run-in and seating by drive stand operation or motoring. One hour of brush run-in is required for each 0.001" of commutator grooving and whether the brushes are seated using a drive stand or by motoring. Refer to each procedure for the amount of time that each procedure requires.

A. Required Equipment

Required Equipment	Range and Accuracy	Source
Contactor	24-28 VDC Coil, 100 A	Figure 15
Control Switch	Single pole, single throw	Figure 15
DC Power Supply or Aircraft Battery or Inverter or Power Supply	24-28 VDC, 100 A (min.)	Figure 15
DC Voltage Regulator (Same)	0-30 VDC	Figure 13
DC Voltmeter (V_1)	0-1 VDC , ± 1% of reading	Figure 13
Generator Drive Stand	10,000 RPM at rated load	Not Illustrated
Generator Load Switch (SW_1)	30 VDC, 500 A	Figure 13
Precision Shunt (SH_1)	1,000 A, 1 VDC	Figure 13
Resistor R_1	5 Ω, 100 W	Figure 15
Resistor R_1, P/N 23081-1810 only	10 Ω, 100 W	Figure 15
Shunt Field Switch (SW_3)	30 VCD, 10 A	Figure 13
Support, "V" Blocks	Sponge Rubber Padding	Not Illustrated
Tachometer or Stroboscope	0-15,000 RPM	Not Illustrated
Variable Load Bank (Same)	30 VDC, 0-500 A	Figure 13
Voltage Regulator Switch (SW_2)	30 VDC, 10 A	Figure 13
DC Voltmeter (V_2)	0-30 VDC, ± 1% of reading	Figure 13

Table 4 - Required Equipment

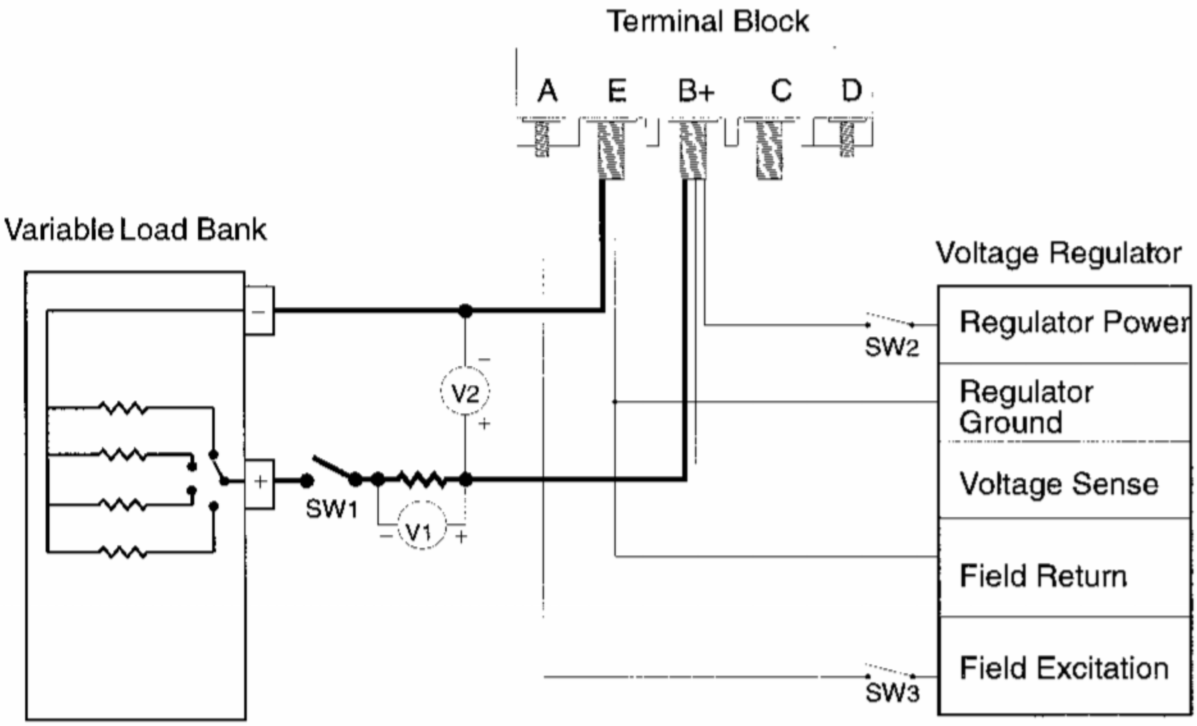

Figure 13 - Run-In by Drive Stand Operation Electrical Connections

B. Final brush run-in by drive stand operation

NOTE: TRW Aeronautical Systems, Lucas Aerospace strongly recommends the drive stand operation method for final brush run-in over the motor operation method.

CAUTION: The unit must be supported at all times during installation onto the drive stand. Do not allow the unit to hang unsupported unless the v-band or the attaching hardware is fully tightened. An undue load on the drive shaft shear section can cause damage to the unit.

(1) Turn OFF all power to the generator drive stand.

(2) Install the generator or starter-generator onto the generator drive stand.

(a) QAD Mount

NOTE: The generator or starter-generator mounts to a mounting adapter plate that is attached to the drive stand. A V-band clamp secures the generator or starter-generator to the mounting adapter plate.

1 Install the applicable spline adapter or mounting adapter plate on the drive stand.

2 While supporting the anti-drive end of the unit, align and install the drive end of the unit to the mounting plate adapter.

3 Make sure the drive stand and the generator or starter-generator mating splines are properly engaged.

4 Install the V-band clamp and tighten the nut to the proper torque.

(b) Direct Mount

NOTE: For direct mount generators and starter-generators, use MIL-L-23699 lubricating oil to provide oil mist lubrication for the drive end ball bearing during brush run-in.

1 Install the applicable spline adapter on the drive stand.

2 While supporting the anti-drive end of the unit, align and install the drive end of the unit to the drive stand mounting pad.

3 Make sure the drive stand and the generator or starter-generator mating splines are properly engaged.

4 Secure the unit to the drive stand as required.

(3) Connect the generator or starter generator to the electrical test circuit.

(a) Turn all power OFF at the generator drive stand.

(b) Connect the generator or starter-generator to the test circuit.

(c) Assemble the terminal block hardware to the terminal block.

(4) For Cored Type and Particle Flake Brushes.

Operate the generator (or starter-generator as a generator) at 8,000 to 10,000 RPM, load per Table 5 and 28 VDC, until the brushes are seated a minimum of 75% in the axial plane and 100% in the rotational plane. (Ref Figure 14)

S/G Full Rating	550A	400A	300A	250A	200A	160A	150A
First Step - 5 min.	200A	150A	100A	100A	75A	75A	50A
Second Step - 10 min.	300A	225A	175A	175A	100A	100A	75A
Third Step - 15 min.*	400A	300A	225A	225A	150A	150A	100A

Table 5 - Cored Type and Particle Flake Brush Values

* If brush seating does not meet requirements of Figure 14, repeat last load step until requirements are met.

NOTE: Table 5 'Third Step' (15 min.) applies only to non-grooved commutators.

NOTE: For a grooved commutator, Table 5 'Third Step' will take approximately 1 hour for particle flake or 4 hours for core type brushes.

NOTE: A properly seated and run-in portion of a brush appears as a smooth, semi-gloss surface. (Ref Figure 14)

(5) For Long Life Brushes.

Operate the generator (or starter-generator as a generator) at 8,000 to 10,000 RPM, load per Table 6 and 28 VDC, until the brushes are seated a minimum of 75% in the axial plane and 100% in the rotational plane. (Ref Figure 14)

Brush P/N	23081 -1810	23064 -1696	23088 -1321	23088 -1322	23079 -1281	23080 -1971	23095 -2480
First Step - 5 min.	75A	100A	100A	100A	100A	150A	200A
Second Step - 10 min.	100A	175A	175A	175A	175A	225A	300A
Third Step - 15 min.*	150A	225A	225A	225A	225A	300A	400A

Table 6 - Long Life Brush Values

* If brush seating does not meet requirements of Figure 14, repeat last load step until requirements are met.

NOTE: During typical brush replacement and run-in, drive stand run-in time is approximately 1 hour for long life brushes.

NOTE: A properly seated and run-in portion of a brush appears as a smooth, semi-gloss surface. (Ref Figure 14)

(6) Turn OFF all power to the drive stand.

(7) Remove the unit from the drive stand.

WARNING: **COMPRESSED AIR USED FOR CLEANING CAN CREATE AIRBORNE PARTICLES THAT MAY ENTER THE EYES. PRESSURE MUST NOT EXCEED 30 PSIG (207 kPa). EYE PROTECTION IS REQUIRED.**

CAUTION: Do not direct the air jet at the bearings to avoid damage to the bearing seals.

CAUTION: Be sure Compressed Air is free of oil, water or any other contaminates.

(8) Blow out the carbon residue from the inside of the unit using compressed air.

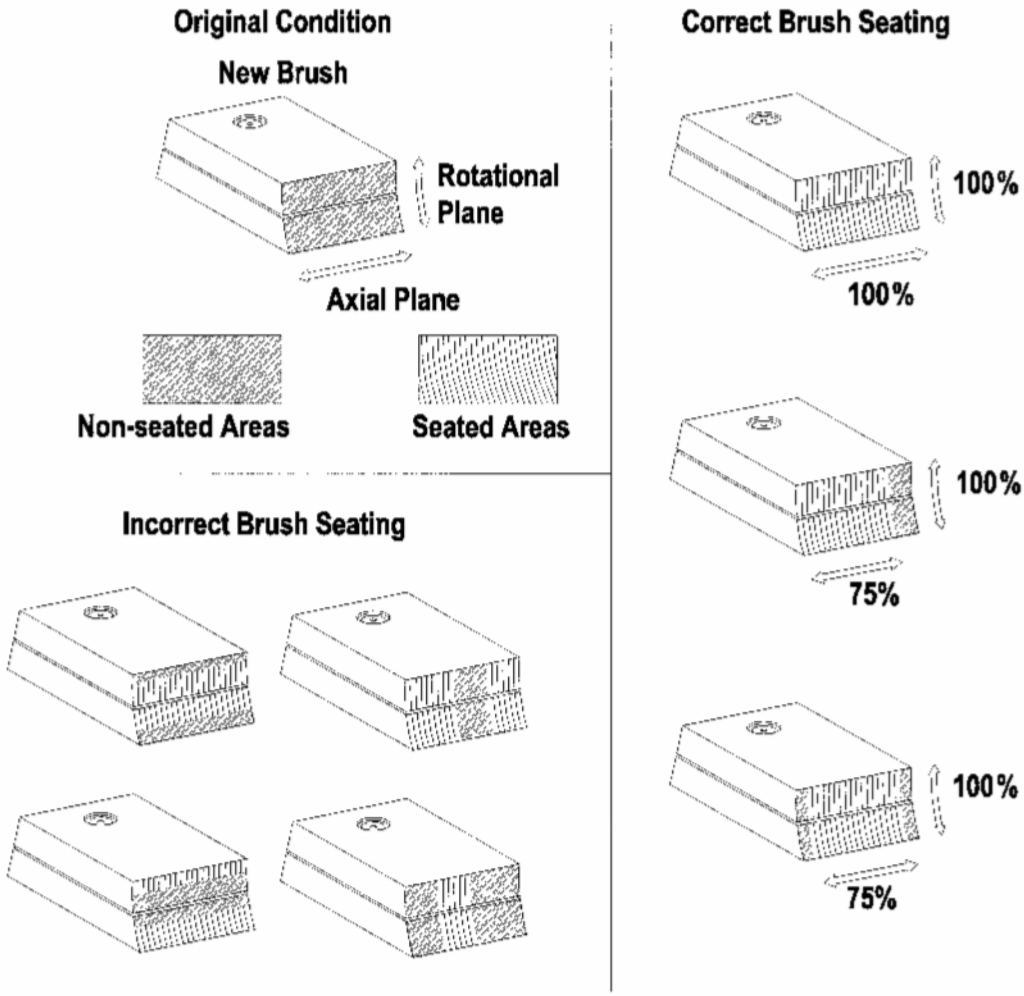

Figure 14 - Proper and Improperly Seated Brushes

C. Final Brush Seating or Run-in by Motoring

NOTE: Brush run-in by motoring is a supplementary procedure and should be used only when equipment limitations prohibit brush run-in by drive stand operation. During typical brush replacement, run-in time by motoring is approximately 12 hours.

NOTE: Long-life brushes cannot be run-in by the motoring method Mount the generator or starter-generator securely on "V" blocks.

CAUTION: The contactor must be open when connecting the power supply.

(1) Make all the electrical connections. (Ref Figure 15)

(2) Close the switch to energize the circuit.

NOTE: If the unit rotates in the direction opposite of normal rotation, check for incorrect internal or external lead connections.

(3) Operate the unit as a motor until the brushes are fine seated - a minimum of 75% in the axial direction and 100% in the rotational Plane.

NOTE: A properly seated and run-in portion of a brush appears as a smooth, semi-gloss surface. (Ref Figure 14)

(4) With the connector open, turn OFF the power supply.

(5) Disconnect the unit from the motoring equipment.

WARNING: COMPRESSED AIR USED FOR CLEANING CAN CREATE AIRBORNE PARTICLES THAT MAY ENTER THE EYES. PRESSURE MUST NOT EXCEED 30 PSIG (207 kPa). EYE PROTECTION IS REQUIRED.

CAUTION: Do not direct the air jet at the bearings to avoid damage to the bearing seals.

CAUTION: Be sure Compressed Air is free of oil, water or any other contaminates.

(6) Blow out the carbon residue from the anti-drive end of the unit using compressed air.

4. Button Up the Unit

Attach the air inlet/fan cover or brush access cover to the unit with the applicable attaching hardware after the completion of brush seating and run-in.

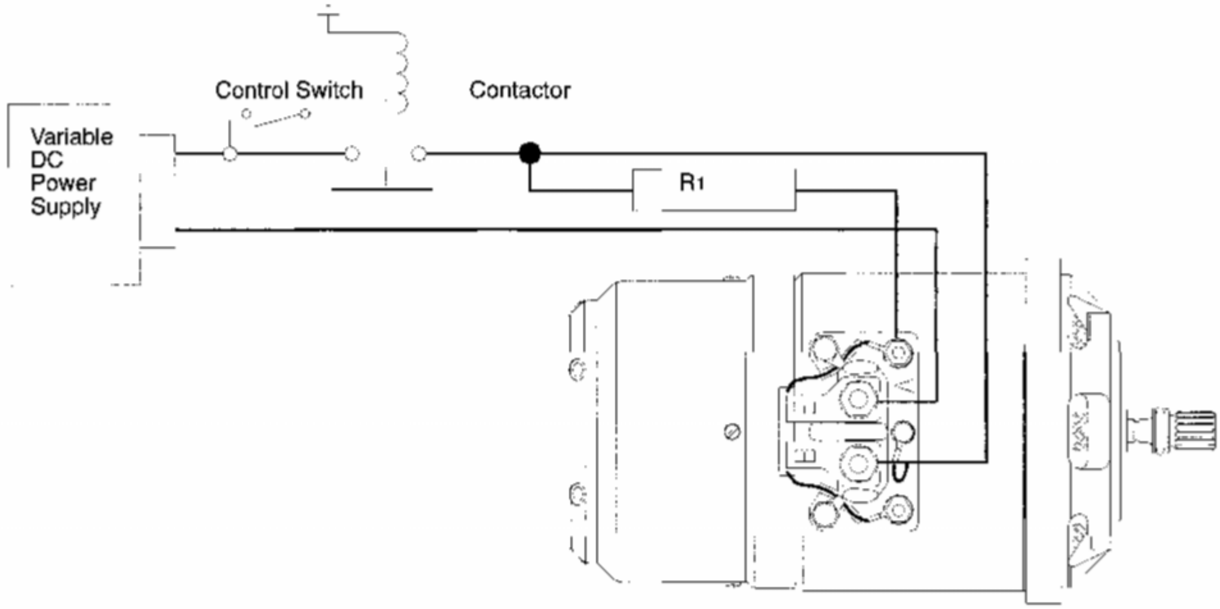

Figure 15 - Run-In by Motoring Electrical Connections

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UNIT INSTALLATION

1. Introduction

This section provides instructions for the installation of the unit onto the aircraft.

2. Tools and Materials

The following materials are required for unit installation.

Tool/Material	Description/Specification	Source
Cleaning pads/wiping cloths	Lint-free, soft fabric	Commercially Available
Isopropyl Alcohol	TT-I-735, Grade A Read **WARNING** and MSDS before using this material	Commercially Available
Plastic or Rubber Mallet	Not Applicable	Commercially Available

Table 7 - Tools and Materials

3. Unit Installation

A. Clean the exterior of the unit

WARNING: ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. DO NOT USE NEAR OPEN FLAMES, WELDING AREAS, OR ON HOT SURFACES. INHALATION OF VAPORS CAN CAUSE DROWSINESS, DIZZINESS, AND HEADACHE. CONTACT WITH SKIN CAN CAUSE IRRITATION. IF HANDLING LARGE QUANTITIES [1 GALLON (3.79 L) OR MORE], USE IN A WELL-VENTILATED AREA

Remove dirt and carbon residue from the drive shaft, the QAD adapter, and the exterior of the unit with a soft, lint-free cloth moistened with isopropyl alcohol.

B. Apply a coating of grease to the drive spline

NOTE: This is not applicable to those units with O-rings.

Apply a coating of the aircraft manufacturer's specified grease to the drive spline and mating accessory pad.

C. Install the unit onto the mating accessory pad

CAUTION: The unit must be supported at all times during installation. Do not allow the unit to hang unsupported unless the v-band or the attaching hardware is fully tightened. An undue load on the drive shaft shear section can cause damage to the unit.

Three common installations are outlined below:

(1) QAD Kit Installation

(a) Align the dowel guide pins in the mounting adapter plate with the guide holes in the drive end end bell.

(b) Press the unit onto the mounting adapter. Support the unit until the V-band clamp is fully tightened.

(c) Position the V-band over the mating flanges.

(d) Latch the T-bolt. Partially tighten the nut on the T-bolt. Continue to support the unit.

(e) Lightly tap around the circumference of the V-band with a plastic or rubber mallet.

(f) Tighten the nut on the T-bolt to 2/3 the recommended torque value stamped on the V-band.

(g) Tap around the circumference of the V-band with a plastic or rubber mallet while tightening the nut on the T-bolt to its full recommended torque value.

(2) Slotted Mounting Holes (Button Holes)

(a) Position the unit so that the clearance holes align with the attaching hardware.

(b) Rotate the unit to engage the slots. Continue to support the unit until the attaching hardware is tightened.

(c) Tighten the attaching hardware.

(3) Round Mounting Holes

(a) Position the unit on the aircraft. Support the unit until the attaching hardware is tightened.

(b) Tighten the attaching hardware.

D. Make Electrical Connections

NOTE: Prior to making electrical connections, be sure that all terminal lugs and contact surfaces are clean.

(1) Make all electrical connections between the aircraft and the unit.

(2) Torque the terminal nuts or bolts to the required torque specified in the aircraft manufacturer's maintenance manual. If values are not provided, use the values in Table 8.

Terminal Size	Torque in-lbs (Newton Meters)
3/8 inch	220-235 (24.88-26.53 N·m)
5/16 inch	125-135 (14.14-15.27 N·m)
#10-32	20-25 (2.26-2.83 N·m)

Table 8 - Torque Limits

(3) Make sure that all electrical connections are correct and attaching hardware is secure.

(4) Install the terminal block cover (where applicable).

E. Perform an Aircraft Electrical System Performance Check

(1) Perform an engine motoring run-in in accordance with the aircraft manufacturer's specifications. Ensure that normal engine speed is attained.

(2) For QAD mounted units, check the V-band attaching hardware for the proper torque after initial engine start and shut down. If necessary, tighten the V-band attaching hardware for proper torque.

(3) After the designated cooling period has elapsed, start the engine. During the starting and generating modes, the aircraft ammeters and voltmeters must indicate stable operation of the DC electrical system.

(4) If any problems occur, shut down the engine.

(5) Check all electrical connections. If problems are found with the electrical system's operation or the generator control unit, remove the unit from the aircraft.

(6) Package and ship the unit to an FAA repair station authorized for accessory overhaul. Include a brief description of the damage condition.

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PACKAGING AND SHIPPING OR STORING THE UNIT

1. Storage

If generator or starter-generator will be stored for more than 12 months, re-test according to the TESTING AND FAULT ISOLATION section of the appropriate CMM before placing it into service.

The following materials are required for packaging the DC generator or starter-generator.

Description	Specification
Tag - 1 (domestic), 2 (overseas)	Commercially available
Grease	MIL-PRF-81322
Chemically Neutral Protective Paper	Commercially available
Polyethylene (Plastic) Wrap	Commercially available
Cardboard Tubing	Commercially available
Waterproof Tape - Pressure Sensitive	Commercially available
Steel or Plastic Bands	Commercially available
Packing Material - shock absorbing foam rubber, styrofoam or bubble wrap	Commercially available
Original Shipping Box or Alternate (Ref. para. 5.(a, b)	Commercially available

Table 9 - Packaging Material

A. General Information

(1) Before preparing generator or starter-generator for shipment or storage, record the following information on tag:

- Model Number
- Serial Number
- MOD Status
- Packing Date

(2) Include applicable documentation with unit:

- Testing
- Repair
- Inspection/Check
- Installation Documentation

B. Packaging

(1) Apply MIL-PRF-81322 lubricating grease to drive spline.

(2) Wrap drive spline with chemically neutral protective paper according to MIL-B-121A, Grade A, Type II.

(3) Attach a section of cardboard tubing over drive shaft.

(4) Wrap generator or starter-generator in chemically neutral protective paper or polyethylene (plastic) wrap and seal with waterproof tape.

(5) Place cushioned well-protected generator into original shipping container (if possible) or make alternate container as follows:

(a) Construct alternate shipping container of double walled corrugated cardboard, minimum 0.625" (16 mm) thick pine board, or equivalent.

(b) Internal dimensions of shipping container must have a minimum of 6" (152 mm) clearance from generator or starter-generator

CAUTION: Make sure unit is protected from sides of shipping container and from movement inside container.

(6) If applicable, place new O-ring, sealed within protective wrap, inside the container. Include instructions to lubricate and install O-ring on drive end of drive shaft prior to installation of unit on aircraft

(7) Put all applicable inspection, test, repair, and installation documents in shipping container with generator or starter-generator

(8) Close shipping container using heavy duty staples.

(9) Apply waterproof tape to all seams of shipping container.

(10) Wrap steel or plastic bands tightly around container to secure.

(11) Tape tag to exterior of container making sure all information is visible.

TOOLS AND FIXTURES

2. General

Use Figure 16 to fabricate a Stator support to be used during brush inspection and replacement.

Fabrication Instructions	Required Material
Unless otherwise specified, dimensions are in inches	Hardwood

Rating (Amp)	A	B	C	D
150 & 200	9.00	4.50	5.375	3.00
300 & 400	9.00	4.50	6.25	3.00
550	10.00	5.00	7.00	3.00

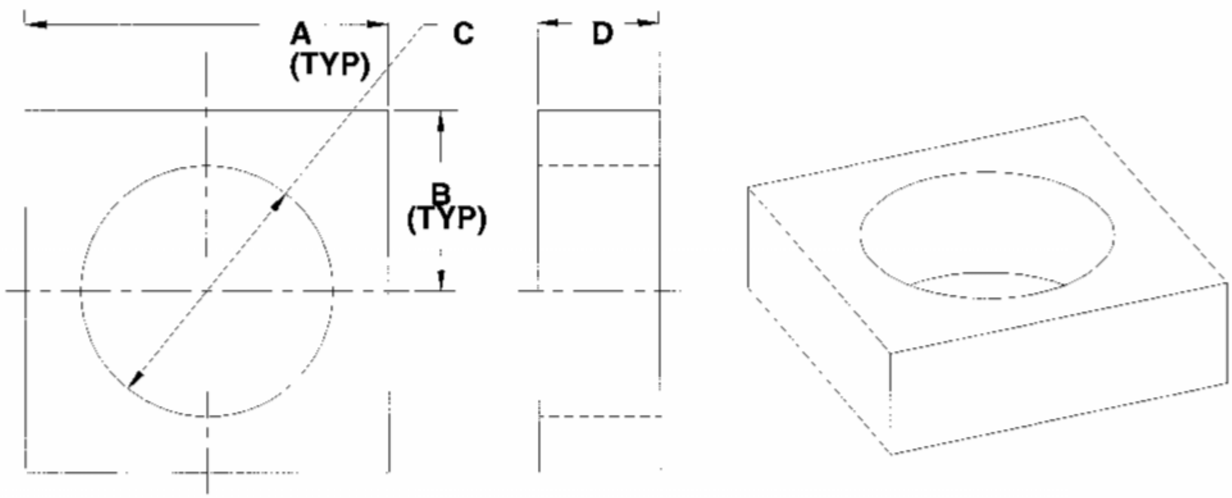

Figure 16 - Stator Support

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